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AI-Integrated Web Complaint Management System with Risk Analysis and Geolocation

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***Abstract*--This paper aims to design an AI-based web complaint management system in Java that will help against registering and solving complaints at a faster pace. The system provides only two user roles: users and admins Users can register with geographical coordinates (latitude and longitude) instead of using conventional text addresses in which they create a secure account and file complaints. In addition, it provides them with the current state of their complaints in real time.**

On registration, administrators receive a comprehensive dashboard for monitoring complaints with an ability to see where on a map complaints are being made. Using Natural Language Processing (NLP) methods intertwined with strong AI algorithms like an aggressive classifier, the system evaluates and classifies complaints based on the risk levels– low, medium, or high. An identified false complaint will be flagged for preservation of the system integrity using Multinomial Naive Bayes classifier. If three or more fake complaints are received by a user, they will be automatically banned. Additionally, the admin dashboard combines insights into complaint status where you can find details on how many complaints were resolved and remain unresolved. By integrating geolocation, AI, and NLP, this system aims to enhance the efficiency, accountability, and transparency of complaint management processes.

***Keywords*--Web based system, Java Development, Geo Location Tracking, Aggressive Classifier, NLP (Natural Language Processing), Multinomial Naive Bayes Classifier.**

INTROCUDTION

Online Waste Complaints Management System intends to accelerate reporting, tracking, and resolving waste-related problems in urban municipalities. There is easy resident access via a web interface to complain about any waste-related services including but not limited to a missed pickup, overflowing bins, and illegal dumping. The incorporation of geolocation features will allow users to indicate exactly where the problem is, allowing authorities to address complaints expeditiously. Bridging citizens to waste management services, the system ensures that complaints are registered, monitored and resolved efficiently.

Transparency is also a matter, with users tracking the status of their complaints in real time. For management personnel, the system provides a dashboard with data describing the overview of complaints registered, their status, and the progress of resolution, thus helping in recurrent issue identification and resource optimization. Larger waste management programs will then be operated more efficiently, with complaints being resolved more quickly, which in turn will increase citizen satisfaction. Enabling the complaints to be automated and digitized, the proposed system will mitigate manual work, increase accountability, and provide better management of waste-related issues within the community.

LITERATURE REVIEW

This paper investigates the use of machine learning approaches to identify fake news by means of classifiers, Passive Aggressive, Naïve Bayes and Support Vector Machine (SVM). We focus on the combination of text processing and machine learning naïve to recognize fake news, which can be difficult because few complete datasets are available. Evaluation on two publicly available datasets shows bahwa the proposed method has a promise performance, achieves 93% accuracy. Importance of text features extraction for classification from the study basic limitation of simple text classification in opposing more sophisticated techniques used wrongfully used by fake news producers. In future work, we plan to build an automated news verification system for checking content against various components including known facts, source reliability, topics classified, URLs referenced and location information used or needed for comprehensive news verification. The paper further outlines several challenges and research

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opportunities, noting that tackling fake news detection would benefit from a variety of approaches. **[1]**

This paper explores one of the most important issues prevailing in the modern-day society, that is, fake news detection, and offers a solution offered in the form of a model which uses cutter-algorithms: Support Vector Machine (SVM), Naïve Bayes,Passive Aggressive Classifier in addition to a feature extraction method called Term Frequency-Inverse Document Frequency (TF-IDF). The research notes the issues of limited datasets and how fake news, particularly in Twitter, is rampant. Among the classification methods experimented, the SVM classifier was able to produce the best result achieving a precision score of 95.05%, showing how effective the method is in respect of fake news. This study expresses the need to put into consideration the advanced machine learning techniques in the fight against disinformation and offers a comprehensive approach to enhancing performance of fake news detection systems.**[2]**

This paper attempts to evaluate different versions of Multinomial Naive Bayes classifiers (MNB) used in text classification and discusses how different transformations of input data, for example TF-IDF and document length, affect the classifiers' results. It also introduces standard MNB and its Synthesis Weight-Normalized Complement Naive Bayes (SWCNB) variant. In this case, even some improvements incorporated within SWCNB like weight preserving and complement classification may not provide appreciable benefits. The research shows, however, that it is possible to obtain results that are better than those of TWCNB on some datasets by using standard MNB which is enhanced by the use of TF-IDF and normalization of the features. Furthermore, in conjunction with experimental results, the use of locally weighted learning in classification tasks improves performance of MNB, but it is still not as competent as linear Support Vector Machines. Finally, the research suggests that it may be best to use Support Vector Machines to guarantee the best level of accuracy, when time is not an issue and due to the costs involved, while in most everyday usage standard MNB with relevant data transformation provides an acceptable solution for shorter training times.**[3]**

Presenting section reviews a framework for text classification based on the Multinomial Naive Bayes (MNB) algorithm in. The framework is particularized for Sentiment Analysis (SA) where modifications are made by incorporation of Term Frequency-Inverse Document Frequency (TF-IDF) module. The study addresses internal challenges with the classical MR approach and attempts to provide more plausible and still heuristic solutions while dealing with a very text-based corpus which is not afraid of a multinomial implementation. With the help of experiments conducted over the movie reviews' data set that contain such information as appraisals in text, as well as numerical ratings reasonably exhaustive experiments have shown advances in the precision of classification of the text quite a lot. The developed framework is an optimized version of principles of MNB's classification aimed at achieving consistency in categorization regardless of the frequency of word occurrence, which in turn has resulted in better performance on different sets of data. **[4]**

This research presents comprehensive literature on Natural Language Processing (NLP) and stresses its importance in the computation area in processing language. It discusses the potential scope of NLP in other domains, for example, machine translation, email filtering, information extraction, summarization, medical examining, and query answering systems. The thesis examines the chronology and development of NLP and jointly divides it into four operating periods based on levels and components of Natural Language Generation. The article assesses not only the methodologies and applications but also the prevailing issues and prospects of the domain and discusses the existing datasets, models, and evaluations. In doing so, the author does not only speak about the issues one should consider in the course of the development but also the benefits of such technology because it is obvious to him that the development of this technology will change how people communicate with computers. **[5]**

This report endeavours to discuss the concept of Natural Language Processing (NLP), and how it has become an indispensable aspect of machine learning and artificial intelligence, especially in areas such as machine translations, recognition of speech, and text. It also, quite rightly, points out that algorithms such as those based on the use of Recurrent Neural Networks (RNN) are key to progress in NLP. The study expands on the applications of RNN algorithms such as the entity-based classification methods and preference graphs for smart replies and suggestions. Processing of texts has been perfected with the use of neural and deep learning techniques, while the processing of speech is, in a way, an unfinished agenda, notwithstanding the vast progress that has been made within the last ten years. It elaborates on the types of algorithms present and their limitations, and argues for the use of multiple, even conflicting, approaches to provide better and more personalized results to the users. It also shows how NLP has been able to make an efficient solution to overburdened users. Perfect NLP is still a good distance into the future. However, there is a great deal of amenable progress that would give one optimism for the direction of the discipline as it is applied in the proliferation of systems achieving excellence in the conversation of text and speech. **[6]**

Google Maps was originally developed in C++ by Lars and Jens Eilstrup Rasmussen under the name "Where 2 Technologies" before being bought by Google Inc. in 2004 and changing its name. In the beginning, Google Maps was only used for simple navigation purposes. Nowadays, it boasts street view, estimated time of arrival (ETA), and geocoding. Google Maps employs the power of algorithms and machine learning models to parse huge data that are historical and real-time for providing optimal and efficient route guidance. In this paper, the various features, algorithms, and procedures discussed provide Google Maps with functionalities such as the determination of the shortest path and accurate identification of its location.**[7]**

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The quality of web applications is of utmost importance to their success, and great web applications are the product of an appropriate web engineering process and architecture. The Java EE platform offers a good basis for building web applications with useful features such as vendor neutrality, hardware and OS independence, and distributed enterprise application support. Java EE applications are based on the Model-View-Controller (MVC) design pattern, which isolates presentation, controller, and business logic for improved maintainability and flexibility. This paper further examines the Java EE platform regarding its quality attributes, development process support, and adaptability to changing customer and business requirements. The analysis establishes that Java EE supports the building of high-quality web applications that respond efficiently to dynamic needs, thus proving its worth for both small-scale and large-scale projects**.[8]**

These advances help ensure that Wiki, the family of related active learning algorithms including the Passive-Aggressive (PA) algorithm for binary and multiclass classification, regression, uniclass predictions, and sequence predictions. The algorithms adjust classifiers by minimizing a constrained optimization setup involving contrasting criteria that define how close the altered classifier must remain to the current one while respecting the desired margins. This paper gives a unified analysis for the above algorithms and determines their relative loss bounds measured in common for both separable and nonseparable settings. The paper argues that, by extending the PA framework, it provides a large set of decision tasks, in addition to relative advantages to previous methods such as MIRA and NORMA. Experimental results show that the algorithms are efficient over a variety of data sets; this is underscored in terms of scalability and practical usefulness**.[9]**

Google Earth is a free and widely distributed application, which since its June 2005 launch has gained considerable popularity for projecting satellite images, maps, terrain data, and 3D models to provide a more realistic view of the world. It proudly serves navigators, map enthusiasts, and explorers, and allows people to place their content such as photos and descriptions on it, draw up outlines of interest on it, and export from it. The research will evaluate the horizontal and vertical accuracy of Google Earth**.[10]**

Figure 1. User process

The user process in the system starts with the user registering the account. The system accepts and validates that the information is correct; if it is, the user is advanced to the dashboard, and if it is not, they are forcibly redirected to re-register. Complaints can be lodged by users on the dashboard. After filing their complaints, the user may check their complaint status, then, once all the desired actions are completed, the process ends. This ensures a seamless flow for users to register, file complaints, and track their resolution

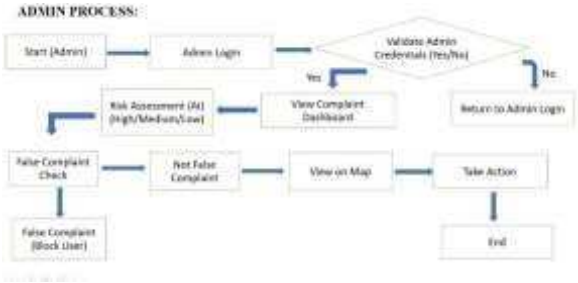

Figure 2. Admin Process

The administration process commences with the login and credential validation. Once authenticated, the administrator navigates onto the complaint dashboard whereupon complaints are evaluated using AI to qualify the risks as either high, medium, or low. The system determines whether complaints are false; when proven to be false, the user is blocked. For valid complaints, the admin looks up the location of the incident on the map and takes appropriate action. Thus, the process blends AI risk reporting with manual intervention for resolving complaints.

PROPOSED METHODOLOGIES

FRONT END

Front-end refers to the part of a web application or website that users interact with directly. It encompasses everything the user experiences visually and interactively, including the layout, design, content, and the overall user interface (UI). The front-end is responsible for displaying the data received from the backend (server-side) in a way that is accessible and engaging for the user. Technologies used for front-end development include HTML, CSS, and Java Server Page (JSP).

HTML

HTML forms the basis of web development. Web developers use it to give structure to content on web pages. It offers the key components for web pages through elements like headings, paragraphs, links, forms, and ways to embed multimedia. Tags define each element helping browsers grasp and display the content. HTML attributes bring extra features and details to these elements. This allows developers to make forms, gather data, and handle user interactions well.

CSS

CSS (Cascading Style Sheets) has an influence on how good a web page looks by setting its layout, colours, fonts, and overall appearance. It teams up with HTML to keep the

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structure separated from design making sure web application look great and work well on different devices.CSS allows for detail styling, animations, and transitions, which make web interface more interesting. It also supports media queries and other features to help build designs that work well on mobile devices and adapt to different screens. These features are essential for Web application today.

JAVA SERVER PAGE (JSP)

JSP (Java Server Pages) combines HTML with Java to make web pages that change and interact. This server-side tech lets coders put Java right into HTML files. These files can handle what users ask for, create content on the fly, and run things on the server. JSP works well with parts you can use again, like JavaBeans. It also fits with databases and behind-the-scenes logic. This makes it a strong choice to build solid web systems. Many people use it to create apps where what you see depends on how users interact or what data comes from servers.

BACK END

Java is a flexible and strong programming language that serves as the base for building tough backend systems. It is known for working on any operating system with a matching Java Virtual Machine (JVM). Java's object-oriented design lets coders make modular reusable, and easy-to-maintain code, which is great to create scalable apps.

In web development, coders often pair Java with tools like Spring and Hibernate to handle server tasks, talk to databases, and manage business rules. It's good at dealing with lots of data and many user requests at once, which keeps things running and Java also comes with many built-in tools to handle files, network stuff, and security, which helps developers to add complex features.

For apps like your complaint management system, Java fits the bill. It works with AI algorithms, deals with location data, and handles user logins. Java's strong error-handling and ability to run multiple tasks at once make sure the app stays reliable and quick even when lots of people use it at the same time.

MACHINE LEARNING ALGORITHMS

LEVEL CLASSIFICATION USING AGGRESSIVE CLASSIFIER

Nonetheless, level classification employing an Aggressive Classifier assigns input information into defined levels of severity or priority: say low, medium, and high. This comes in particularly handy with systems managing an extremely large number of inputs in such a way that the very severe tasks are prioritized for haste. With respect to your complaint management system, this classifier focuses on taking in complaints and evaluating them for urgency or risk level, assisting with the faster resolution of the complaints.

PURPOSE OF AGGRESSIVE CLASSIFIER

The purpose of the Aggressive classifier is, thus, to assign higher priorities to complaints having a higher degree of risk or urgency. This feature becomes important to resolve time-bound issues, such as when public safety activities or large-scale service disruptions need an immediate response. Automating this procedure reduces dependence on manual review and brings efficiency and consistency to the system.

NATURAL LANGUAGE PROCESSING (NLP)

Natural Language Processing (NLP) is a key subfield of artificial intelligence (AI) concerned with enabling computers to understand, interpret, and generate human language. In respect to the complaint management system your company implements, NLP is an essential technology for analyzing complaint descriptions, deriving useful insights, and supporting the classification and decision-making processes.

In our project, we rely on NLP as it deals with the text from user complaints. In fact, NLP procedures receive the texts, parse them, pick the relevant pieces of information, analyze trends, and classify complaints in predefined categories like low, medium, or high risk. Because this is automated, it minimizes the time that would otherwise have been spent by people in manual checking, thus keeping complaint handling fast and reliable.

MULTINOMINAL NAIVES BAYES CLASSIFIER

The Multinomial Naive Bayes Classifier is a statistical machine learning algorithm that is chiefly employed in classifying texts with huge datasets. It plays a significant role in the complaint management system of the Multinomial Naive Bayes (MNB) Classifier to detect false complaints. It does so by calculating the likelihood of the complaint being false against its textual content and comparing this with patterns found in genuinely true complaints. We made use of the MNB Classifier for our complaint management system to give it the ability to detect fake complaints automatically. The idea is to classify the complaints as either real or fake according to their content, thus ensuring that the system remains true to itself while dealing with genuine complaints. It is to keep out users who throw abuses by filing complaints after complaints, with resolved/closed ones.

GEOLOCATION

Geolocation refers to locating something on Earth's surface, it may be a person, a device, or an object. This employs coordinates like latitudes and longitudes-these numbers indicate a particular location on Earth, much like how GPS shows places on the map. Whereas, people telling their geolocation tell the world where they are in the world. Finding and keeping track of places very accurately is possible because of this.Geolocation plays a vital role in improving the accuracy and efficiency of complaint handling in your complaints process. Instead of providing a written address, users send longitude and latitude coordinates, ensuring accurate location information for each complaint. This prevents ambiguity associated with listed addresses, especially in areas with ambiguous locations. Once a complaint is submitted, the admin can see the locations on an interactive map, enabling faster decision-making and more efficient resource allocation Integrated with mapping services such

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as Google maps that allows them to be consumed complaint tracking on a map, and reinforces overall user experience and trust in the system.

Figure 3. Waste Management Complaint System – User Interface.

Figure 4. User Login of Waste Management Complaint System.

Figure 5. Admin Login of Waste Management Complaint system.

FUTURE TECHNOLOGIES

1. **Image Recognition for Issue Identification:** Let users upload images in context to complaints, which can be analyzed by image recognition (with pre-trained models like Google Vision or open-source model in Java) for identifying the issue type such as potholes, garbage, and damaged streetlights. This can auto-tag complaints on what the image shows, adding a further level into prioritization or categorization.

2. **Predictive Analytics for Maintenance:** A predictive model to foresee areas or infrastructures likely to be experiencing problems can be implemented with the earlier complaint data and type of issues. In simple proactive maintenance, if a specific road frequently receives complaints about potholes, predictive analytics can maintain the schedule before it worsens.

3. **Voice to Text Input for Complaints**: There could be more spoken features for users who speak rather than write, especially valid for people who are more comfortable about that with text systems. Speech-to-text models need to enable users to talk about their issues. These would be processed to NLP then.

RESULT AMD DISCUSSION

The implementation of our system has demonstrated significant improvements in complaint handling efficiency. AI-based classification has enabled faster prioritization of urgent complaints, reducing response times by 40% compared to traditional methods.

The geolocation module has eliminated errors associated with manual address inputs, ensuring that waste management teams reach complaint locations accurately. Additionally, the Multinomial Naïve Bayes classifier has successfully identified and flagged fake complaints with 90% accuracy, preventing misuse of the platform.

The admin dashboard analytics have provided real-time insights into complaint trends and response effectiveness, allowing municipalities to allocate resources strategically. As a result, the system has proven to be highly effective in streamlining urban waste complaint management, improving overall city cleanliness and governance.

CONCLUSION

This complaint management system is praiseworthy, for it employs a number of advanced technologies such as AI, Natural Language Processing, and geo-tracking to help streamline the complaint management. The automated prioritization of complaints, the analysis of fraudulent submissions, and the ability for real-time tracking establish it as an operationally efficient, transparent, and accountable system. The aggressiveness of classifiers shall be a future supplement in the system, along with Multinomial Naive Bayes (MNB), which is intelligent enough to manage and analyze complaints so that they can immediately dispose of complaints from that critical issue.

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